11. Which of the following is an electromagnetic wave?
(a) α - rays (b) β - rays
(c) γ - rays (d) all of them

12. Which one of them is used to produce a propagating electromagnetic wave?
(a) an accelerating charge
(b) a charge moving with constant velocity (c) a stationary charge
(d) an uncharged particle

13. If $E = E_o \sin[10^6 x - \omega t]$ be the electric field of a plane electromagnetic wave, the value of ω is
(a) 0.3×10^{-14} rad s^{-1}
(b) 3×10^{-14} rad s^{-1}
(c) 0.3×10^{14} rad s^{-1}
(d) 3×10^{14} rad s^{-1}

14. Which of the following is NOT true for electromagnetic waves?.
(a) it transports energy
(b) it transports momentum
(c) it transports angular momentum
(d) in vacuum, it travels with different speeds which depend on their frequency

15. The electric and magnetic fields of an electromagnetic wave are
(a) in phase and perpendicular to each other
(b) out of phase and not perpendicular to each other
(c) in phase and not perpendicular to each other
(d) out of phase and perpendicular to each other

Answers

1) b	**2)** d	**3)** d	**4)** c	**5)** a
6) b	**7)** a	**8)** c	**9)** b	**10)** b
11) c	**12)** a	**13)** d	**14)** d	**15)** a

II Short answer questions

1. What is displacement current?
2. What are electromagnetic waves?
3. Write down the integral form of modified Ampere's circuital law.
4. Write notes on Gauss' law in magnetism.
5. Give two uses each of (i) IR radiation, (ii) Microwaves and (iii) UV radiation.
6. What are Fraunhofer lines? How are they useful in the identification of elements present in the Sun?
7. Write notes on Ampere-Maxwell law.
8. Why are e.m. waves non-mechanical?

III Long answer questions

1. Write down Maxwell equations in integral form.
2. Write short notes on (a) microwave (b) X-ray (c) radio waves (d) visible spectrum
3. Discuss the Hertz experiment.
4. Explain the Maxwell's modification of Ampere's circuital law.
5. Explain the importance of Maxwell's correction.
6. Write down the properties of electromagnetic waves.
7. Discuss the source of electromagnetic waves.
8. Explain the types of emission spectrum.
9. Explain the types of absorption spectrum.

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